Excess insulin:

- An insulin-secreting pancreatic tumour (an *insulinoma*). A relative insulin excess more commonly results from a mismatch between carbohydrate intake and the dose of injected insulin in a diabetic individual, e g. when a meal is missed.
- The significant outcome is a reduction in plasma glucose levels (*hypoglycaemia*). Low blood glucose directly activates the sympathetic nervous system leading to tremor, sweating and an increased heart rate.
- The patient become anxious or aggressive before hypoglycaemic coma results. This must be reversed using i/v glucose or glucagon if permanent brain damage is to be avoided.

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Leptin H.
(Voice of fat, appetite regulating H.)

- Leptos = thin, it decrease feed intake at hypothalamic level.
- It is a small protein H secreted from adipose tissue, placenta, ovary and liver.
- It is an autocrine, paracrine and endocrine H.
- It is secreted mainly from perirenal fat and epidydimal fat.

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- **Effects of Leptin:-**

A. Regulate body fat stores through:- Inhibit feeding pattern & Stimulate sympathetic N.S. and metabolism.

B. It stimulate liver cells and skeletal M. to increase oxidation of Fatty A in mitochondria so result in decreased fat content in hepatocytes and muscle cells.

- **So** It increase Oxygen consumption and increase body temperature (regulate energy burning process) in synergism with thyroxin.

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Effect on reproduction:-

- Leptin is the messenger which convey information regarding energy balance to the neuro-endocrine system regulating reproduction. i.e when body fat stores reaches a certain critical amount then leptin will be secreted in enough amounts to start LHRH secretion and activation of hypothalamic-pituitary-gonadal axis (trigger puberty).
- When body fat falls below the threshold level as in anorexia nervosa or in athletes, the ovarian cycle stops.
- ***So, females reach puberty when they attain certain weight not certain age!*** Starvation or fasting result in decreased leptin levels and decreased estradiol levels.

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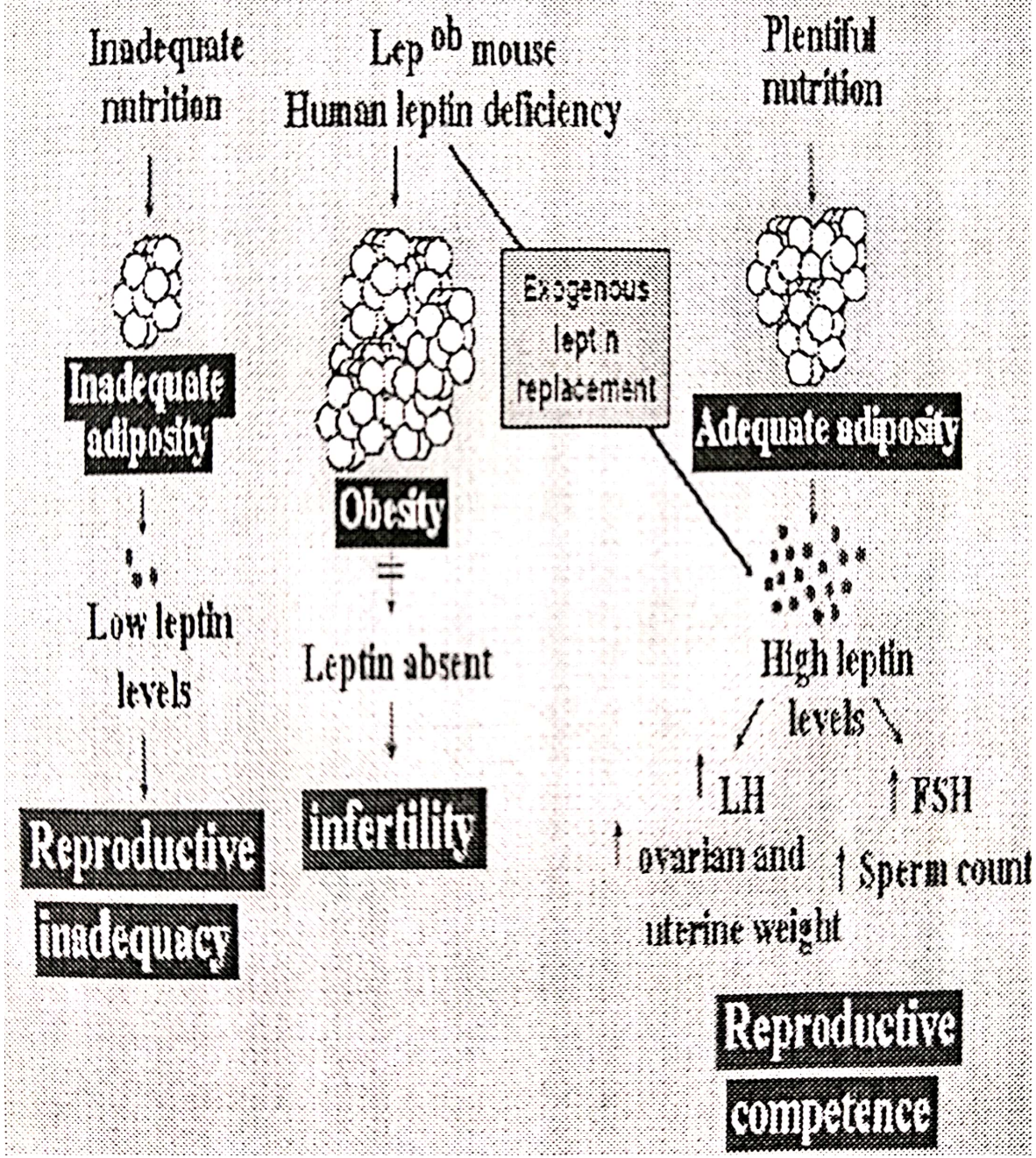
The Critical Fat Mass Hypothesis Revisited



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- It stimulates Growth H. and prolactin H secretion and lower somatostatin H.
- Congenital lack of leptin results in early onset of Obesity, hyperphagia and hypogonadism which respond to Gn RH treatment.
- Obesity also results when Ob-R receptors (Leptin receptors) are insensitive to leptin , (same as type 2 diabetes), so leptin receptor deficient mice suffer from hypogonadism which will not respond to Gn RH treatment.
- **N.B.** Adipose tissue secret more than 15 biological materials among them is Leptin and Resistin hormone.

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- Adipose tissue also contain *aromatase* enzyme which cause excessive formation of estrogens (block secretion of FSH) and excess testosterone causing hirsutism (Male characters in females).
- High levels of LH produced (due to high leptin level) may cause polycystic ovary syndrome and infertility.
- Leptin level increase before puberty, during luteal phase and during pregnancy, pass in milk and affect the neonate.
- Cold lower leptin secretion so feed intake increases.

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- On mammary gland:- Leptin stimulate mammary epithelial cell proliferation, differentiation and is secreted in milk.
- Other effects:
- It stimulate wound healing, Increase endometrium thickness, Stimulate apoptosis of adipose tissue, Stimulate the immune system (activate W.B.Cs) and promotes gluconeogenesis.

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